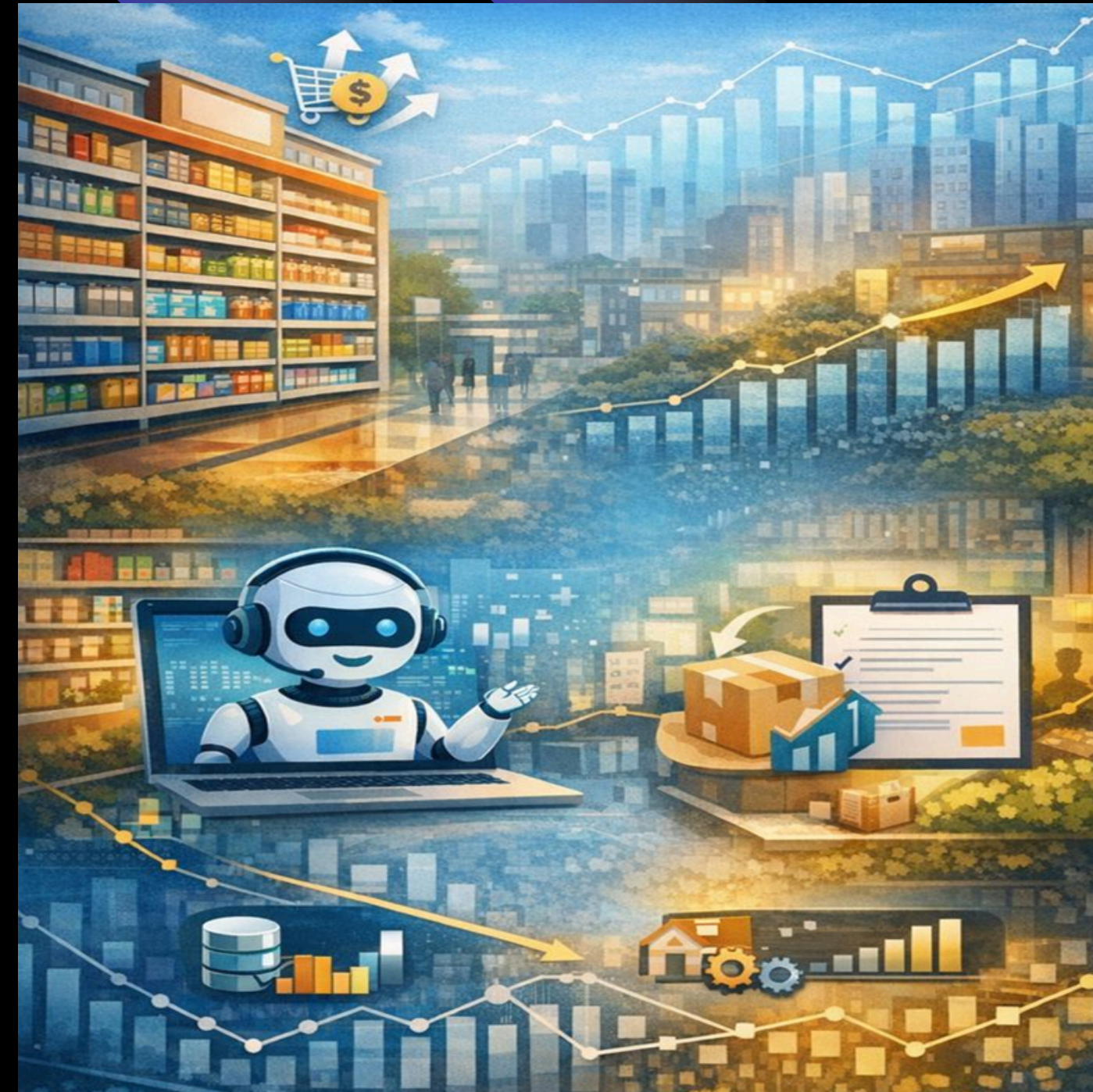


Smart Stock Inventory Optimization for Retail Stores

ENHANCING BUSINESS
EFFICIENCY THROUGH
STREAMLINED STOCKS.

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INTRODUCTION

- In today's competitive retail environment, efficient inventory management is essential for maximizing profitability and customer satisfaction.
- This project introduces a Smart Inventory Optimization System that uses data analytics and machine learning to predict product demand and optimize stock levels.
- This software-based system reduces manual effort, making it a practical and scalable solution for modern retail management.



Problem Statement

1. **Frequent stock-outs**-Lost sales and customer dissatisfaction
2. **Overstocking**- Increased holding costs and product wastage
3. **Manual inventory tracking**- Time-consuming and error-prone
4. **Inaccurate demand forecasting**- Lack of data-driven analysis



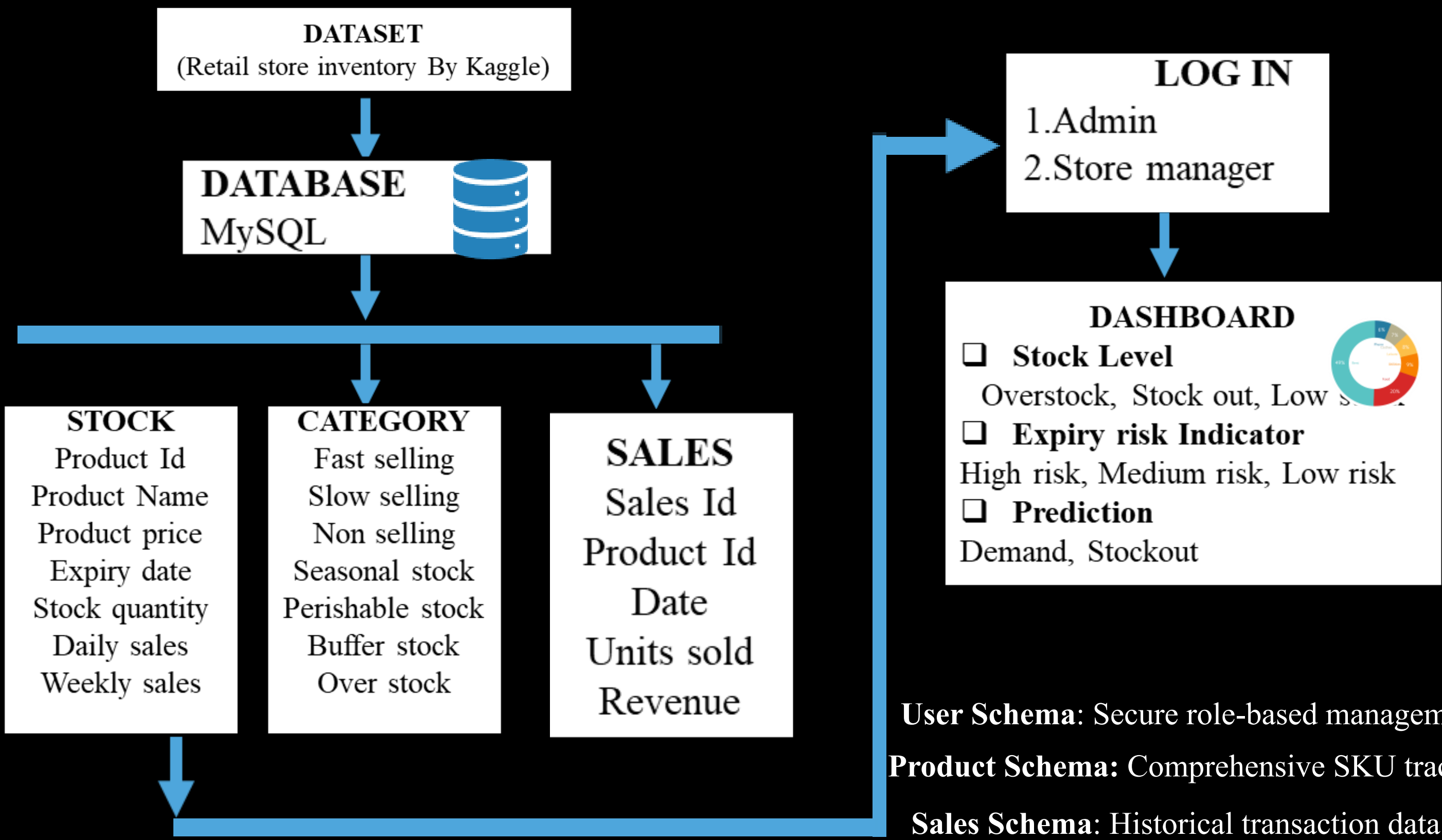
Smart Solutions

- The system continuously monitors inventory levels and applies predefined stock thresholds derived from historical sales data.
- A machine-learning demand forecasting model predicts future sales and identifies potential stock-out risks in advance.
- Real-time dashboard alerts enable proactive restocking, reducing lost sales and ensuring better product availability.



Maintaining the perfect balance in stock levels.

System Architecture

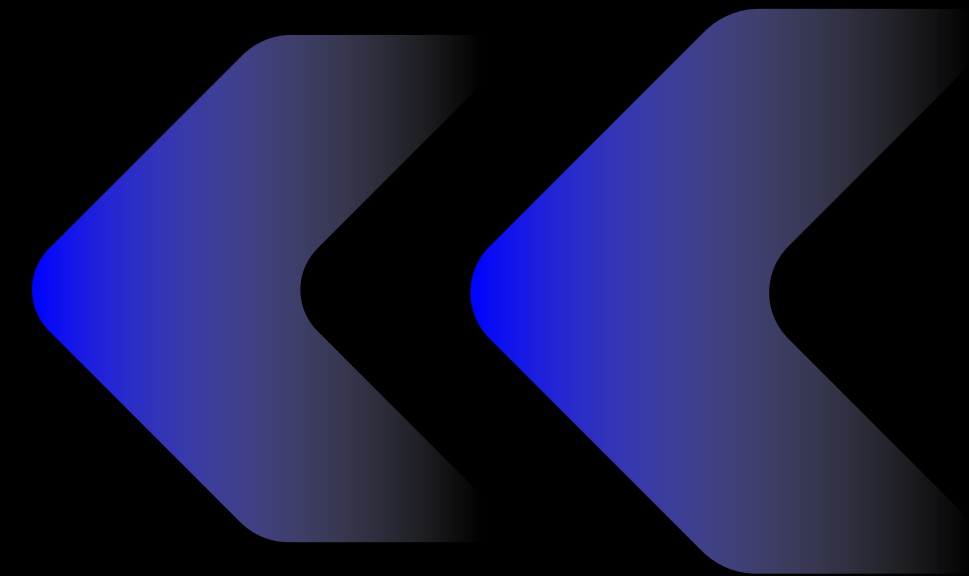


User Schema: Secure role-based management for Admins and Store Managers.

Product Schema: Comprehensive SKU tracking including city and store identifiers.

Sales Schema: Historical transaction data mapped to time-series sequences.

Technology Stack



FRONTEND

Zustand State
Recharts
Efficient Data Fetching
Material UI



BACKEND

Python
FastAPI
SQLAlchemy
Task Scheduler



DATA & AI

Prophet (Forecasting)
MySQL (Database)
Scikit-Learn
NumPy



INTELLIGENCE

XGBoost ML model for
precise time-series sales
forecasting.

Database Schema MySQL

- **Users**- User Id, Role , store_id, Gmail
- **Inventory Data**- id, date , store_id , product_id , category , region, price discount , competitor_pricing ,inventory_level , units_ordered , demand_forecast, holiday_promotion, weather_condition , seasonality, units_sold , city, store_name , product_name
- **Sales Records**- Historical sales data for Model traning.

Optimized storage
for faster order
fulfillment.



Authentication

USER LOGIN

email address

password

☐ Remember Me

LOGIN

Forgot Password?



- **API & Authentication Architecture-**
Backend API Framework: FastAPI



- **Authentication & Security-**
Authentication Service: -
Firebase Authentication



- **Role-Based Access Control**
 1. **Admin Role-** Full access to all application pages
 2. **Store Manager Role-** Restricted access based on assigned store ID

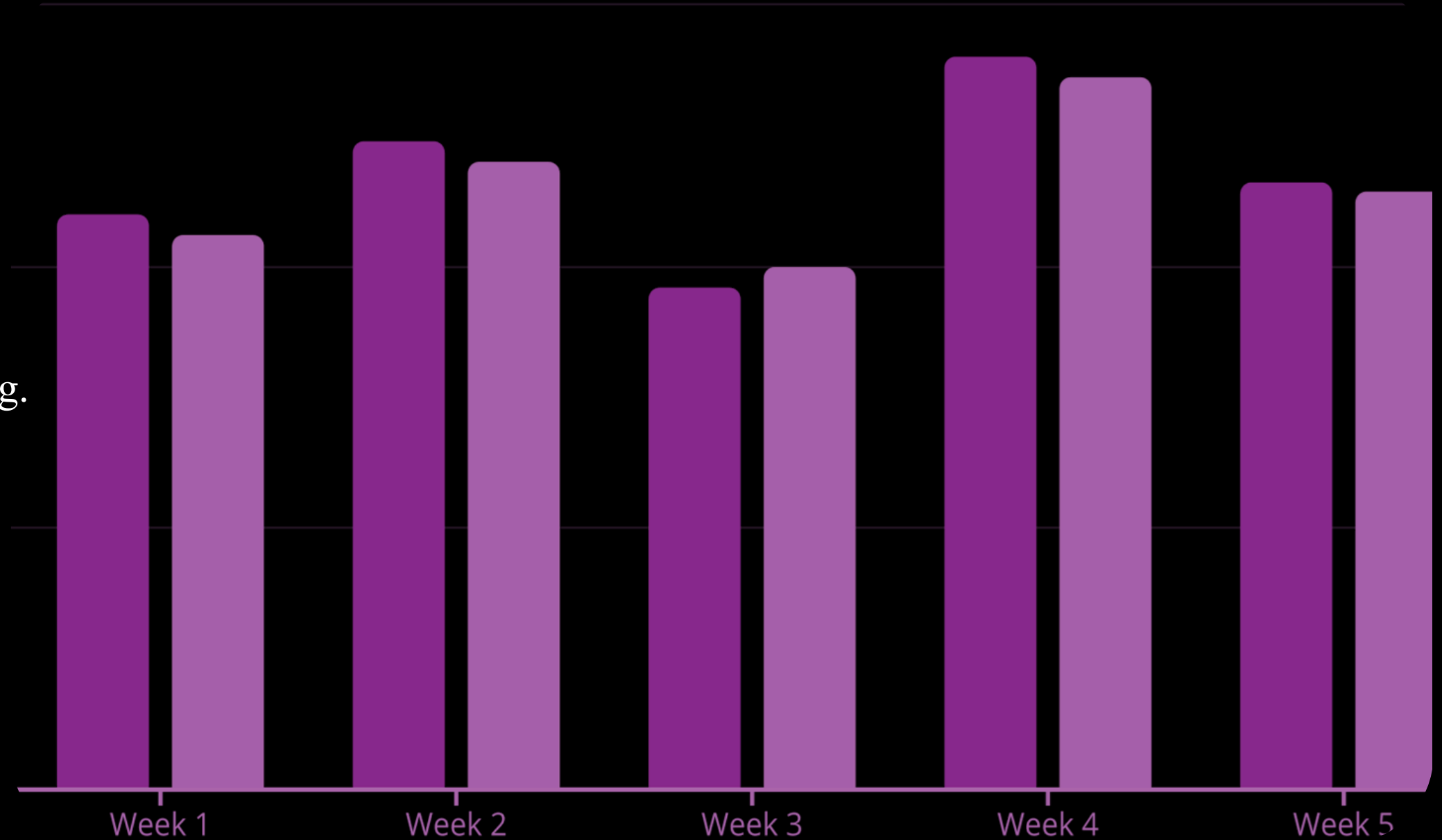
Trained Forecast Model

Model Training: The XGBoost Powerhouse

- `n_estimators=300`: Number of boosting rounds.
- `max_depth=10`: Maximum depth of each tree.
- `learning_rate=0.1`: Step size shrinkage to prevent overfitting.

By leveraging Extreme Gradient Boosting, our system identifies non-linear sales patterns that traditional models miss.

The model processes historical sales volume, seasonality, and localized trends to predict future demand with over 92% accuracy.



INVENTORY ALERTS



- Low Stock Alert
- Empty Stock Alert
- Reorder Alert.

Alert = Predicted Sale > Inventory

- The system suggests reorder for Low and Empty stock

City Dashboard

City Selection

All Stores in City

Unique store Id

Store-wise Sales Data

City Dashboard Insights



Future Scope

Automated Best Supplier
Identification



Integration with Email and
SMS Alerts





**BUILDING A FUTURE OF
EFFICIENT AND
SUSTAINABLE STOCKS.**

**Thank
You..**